Coringtreespotters ms outline

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Introduction

Antropogenic climate change affect many different biological processes. One of the most documented is shifts in phenology—study of recurring life history events. In plants, shifts in spring and autumn phenology have lengthen the window of opportunity during which they can grow. Trees more specifically should grow more with longer growing seasons which is particularly important given how important temperate forests are in offsetting human-emitted greenhouse gas emissions. In these ecosystems, a large portion of that carbon is stored in mature, canopy trees accentuating their critical role in forest carbon sink potential.

However, the long standing that longer growing seasons should increased growth has recently been challenged by studies that failed to find this relationship. Indeed, warmer springs which contribute in large part to extending the growing season have not increased tree growth rate nor annual increment *(**?**). Increased tree growth with longer growing season is an important foundation of current carbon models—crucial tools to help set emission objectives and their related solutions. This highlights the need to understand why trees may not grow more with longer growing seasons.

Wood anatomy and tree growth strategies may guide species growth responses to growing seasons. Wood anatomy in angiosperms tree has a known impact on their growth onset in the spring where ring-porous species start growing several days to weeks earlier than diffuse-porous species. As a result, species with different wood anatomy experience considerably different temperature regime during the growing season. Furthermore, longer growing seasons may benefit slower growth species as appropriate conditions allow them to keep their steady growth for a longer period. In contrast, fast growth species can complete their season growth increment in a short period, making extra days less crucial for them. Additionally, how trees store the carbon that they accumulate (whether it is a direct investment in growth or stored for long term storage) could mute short term responses. Thus, increased carbon storage during the current year that is stored instead of directly invested in growth, highlights the importance of considering multiple years.

The previous year thermal conditions can affect the following year's growth. Deciduous trees are known to use stored carbon at the beginning of the growing season because they need to grow new leaves to start photosynthesis—unlike evergreen trees. Mature trees in particular can rely on carbon that was stored several previous years for their early season growth. This shows that while the environmental conditions during the current year can have a direct effect on growth, it is likely that this poorly understood carbon allocation strategy to storage may be blurred by a multi-year lag effect.

Here, we will look at the growth and growing season length relation ship of mature trees in an urban arboretum. For this, we used 11 species of mature trees across nine years and compared the growing season number of days and the conditions during that period. We also investigated the effect of the previous year thermal conditions on the current years growth. We used growth with tree rings and leaf phenology for the start and end of seasons that was collected by citizen scientists.

Discussion

In this study, we used 49 trees of 11 species to understand if mature trees grew more with longer growing seasons. To do this, we used annual growth with tree rings and leaf phenology for the SOS and EOS. Our findings revealed that warmer and longer growing seasons did not increase growth for any species and decreased growth for one. We also found that the previous year thermal conditions shift growth trends for two species. Finally, most species did not respond to the timing of start and end of season.

While our species share an array of growth strategies, we did not find a consistent growth trend across them. The thermal and calendar season had no effect on any species, except for one. *B. alleghaniensis* growth decrease reveals that this species growth may be inhibited by warmer seasons. Indeed, this negative response to the thermal season is consequential with our observed increase in growth with an earlier end of season—a period where GDD are accumulated at a high rate. Thus, *B. alleghaniensis* benefits from an earlier EOS that leads to fewer GDD accumulation. In addition, *T. americana* grew less with an earlier SOS showing that despite this fast-growing species appear to be negatively impacted by an earlier start of season during—during which days are short and cooler. Ultimately, these findings suggest that there is no difference in the relationship between the conditions and length of the growing season with growth between fast and slow growth species.

Previous year thermal conditions shifted growth only for two *Betula* species. We found diametrically opposed trend between *B. alleghaniensis* and *B. nigra*. *B. alleghaniensis* decreased growth with the current year thermal conditions, but grew more with a warmer previous year, potentially revealing a carbon allocation strategy that prioritize storage before growth. This is expected of slower growth species, though *B. alleghaniensis* is considered a moderately fast-growth species [*REF table]. In opposition, *B. nigra* is fast growth and its increase during the current year only when the previous year conditions were included highlights the importance of considering not only the current year conditions when studying mature tree growth, whatever their growth strategies may be.

Our findings reveal a weak response across species corroborating previous findings on mature trees. Most of our studied did not grow more with a warmer thermal or longer calendar season. However, we note the importance of considering the previous year conditions in future growth and growing season length studies because carbon allocation strategies may blue tree responses to the current year GS length and conditions. While growing seasons are expected to warm and lengthen, whether or not tree growth will follow is still dubious. Whether tree growth stalls or decrease, both will have severe consequences on the extent to which temperate forests will act as a carbon sink that offset human greenhouse gas emissions.

Supplemental results

Table S1: Species level slope parameters which represent ring width change (on the log scale), per scaled predictor (see data analyses section for more details). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Species	GDD	GSL	SOS	EOS
A. rubrum	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
A. saccharum	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
Ae. flava	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
B. alleghaniensis	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
B. nigra	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
C. glabra	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
C. ovata	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
P. deltoides	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
Q. alba	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
Q. rubra	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
T. americana	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

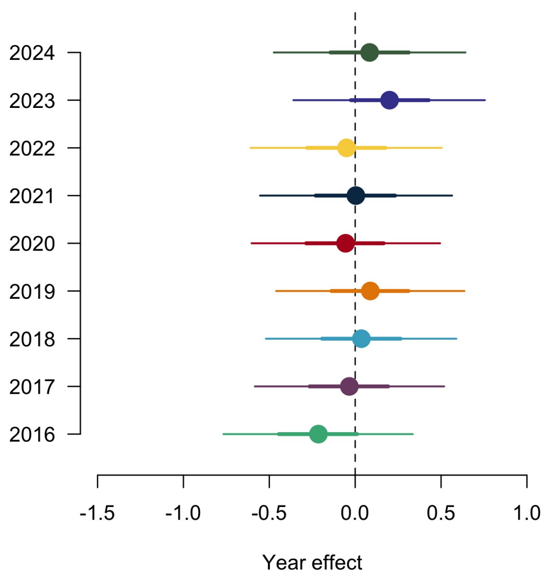

Figure S1: Ring width (on the log scale) deviations from the grand mean for each year using the GDD model (Table S2). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

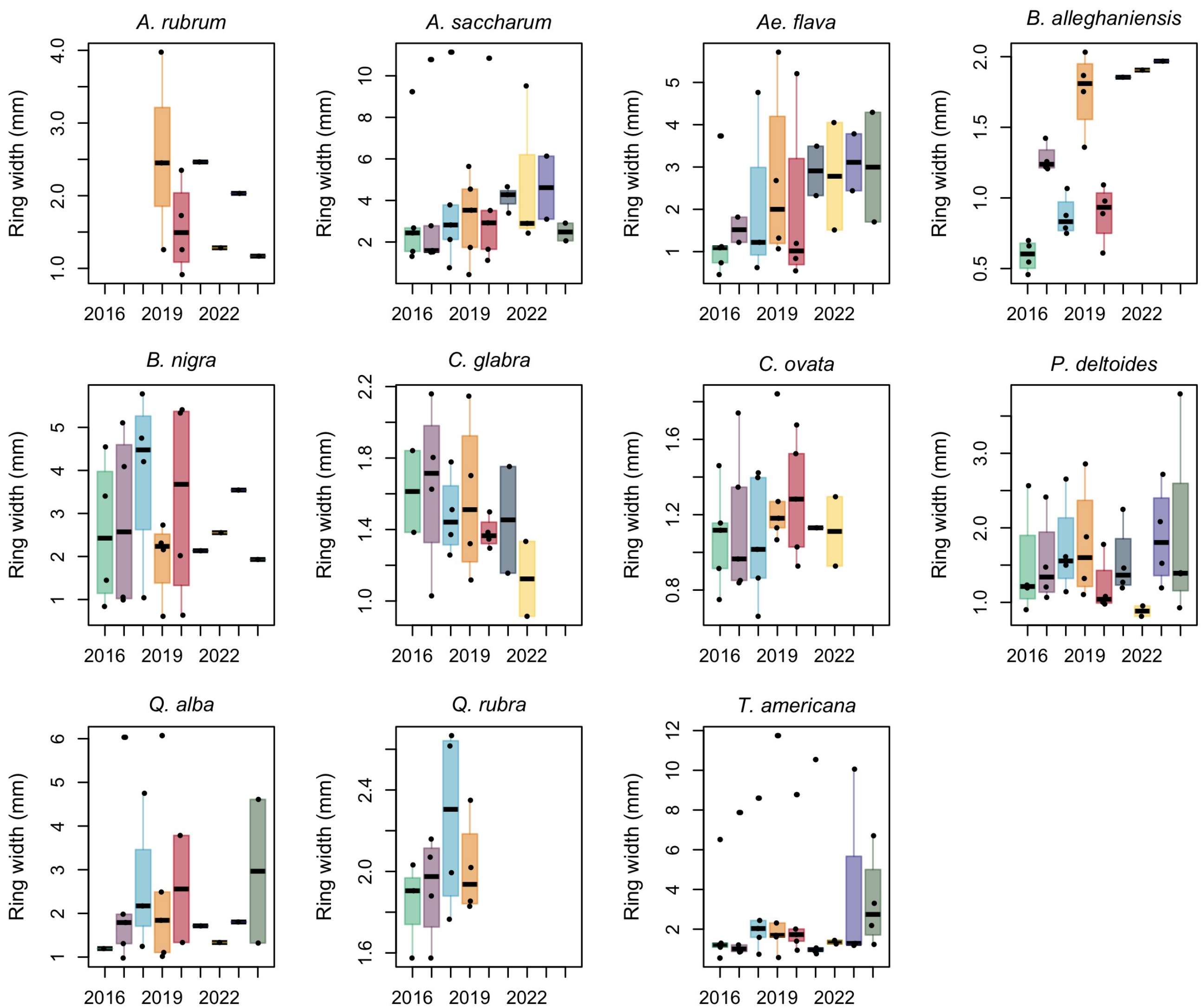

Figure S2: Boxplot representing the empirical data of our ring widths (mm) across years, facetted by species. The dots represent the raw data, the line the mean and the box the interquantile range (IQR).

Table S2: Summary output of each parameter from our four predictors, using ring width (on the log scale) as the reponse variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See data analyses section for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{spp_{A.rubrum}}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{spp_{A.saccharum}}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{spp_{Ae.flava}}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{spp_{B.alleghaniensis}}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{spp_{B.nigra}}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{spp_{C.glabra}}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{spp_{C.ovata}}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{spp_{P.deltoides}}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{spp_{Q.alba}}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{spp_{Q.rubra}}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{spp_{T.americana}}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year_{2019}}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{year_{2020}}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{year_{2021}}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{year_{2022}}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{year_{2023}}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{year_{2024}}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]
$\alpha_{year_{2016}}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{year_{2017}}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{year_{2018}}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{spp_{A.rubrum}}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{spp_{A.saccharum}}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{spp_{Ae.flava}}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{spp_{B.alleghaniensis}}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{spp_{B.nigra}}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{spp_{C.glabra}}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{spp_{C.ovata}}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{spp_{P.deltoides}}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{spp_{Q.alba}}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{spp_{Q.rubra}}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{spp_{T.americana}}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]

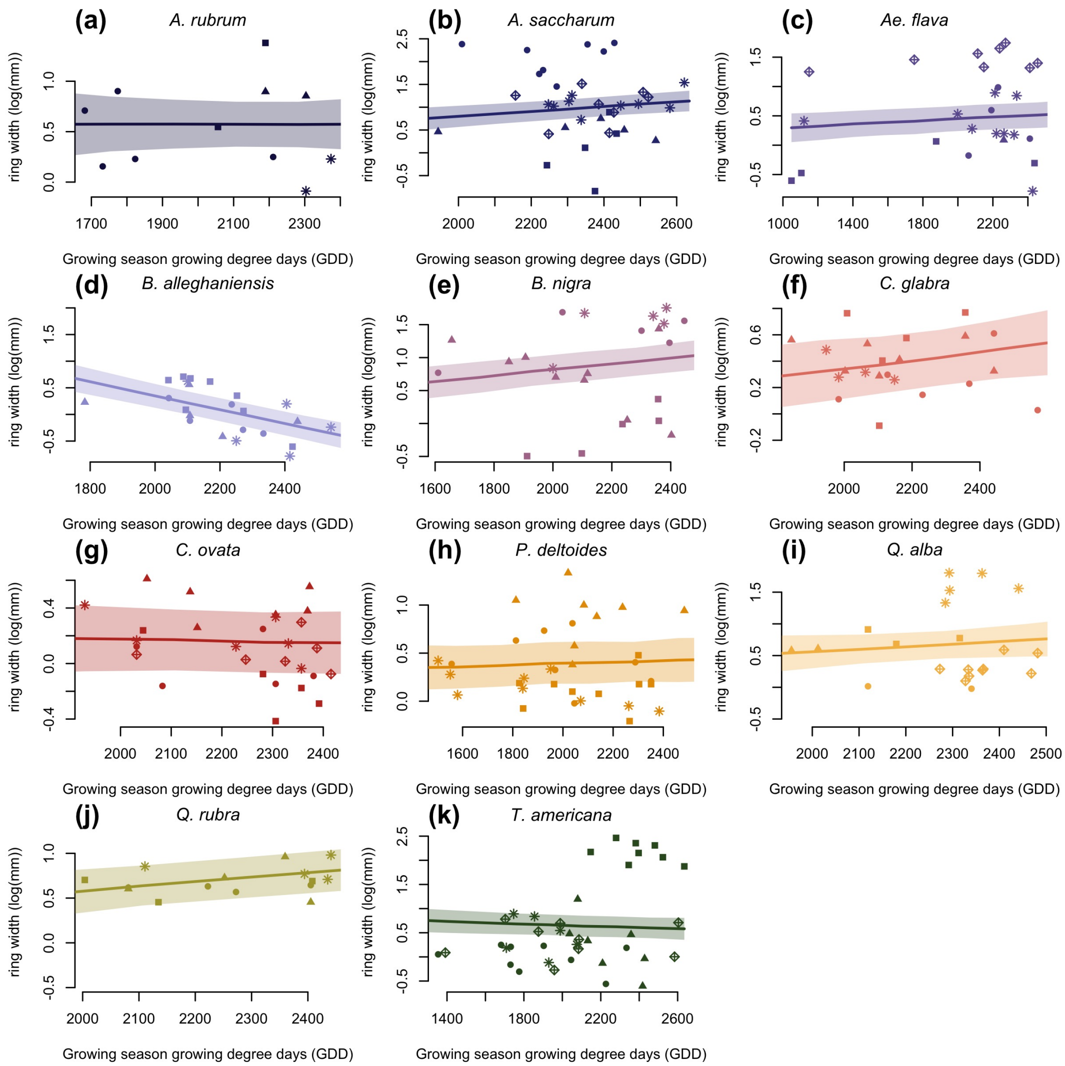

Figure S3: Ring width (on the log scale) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

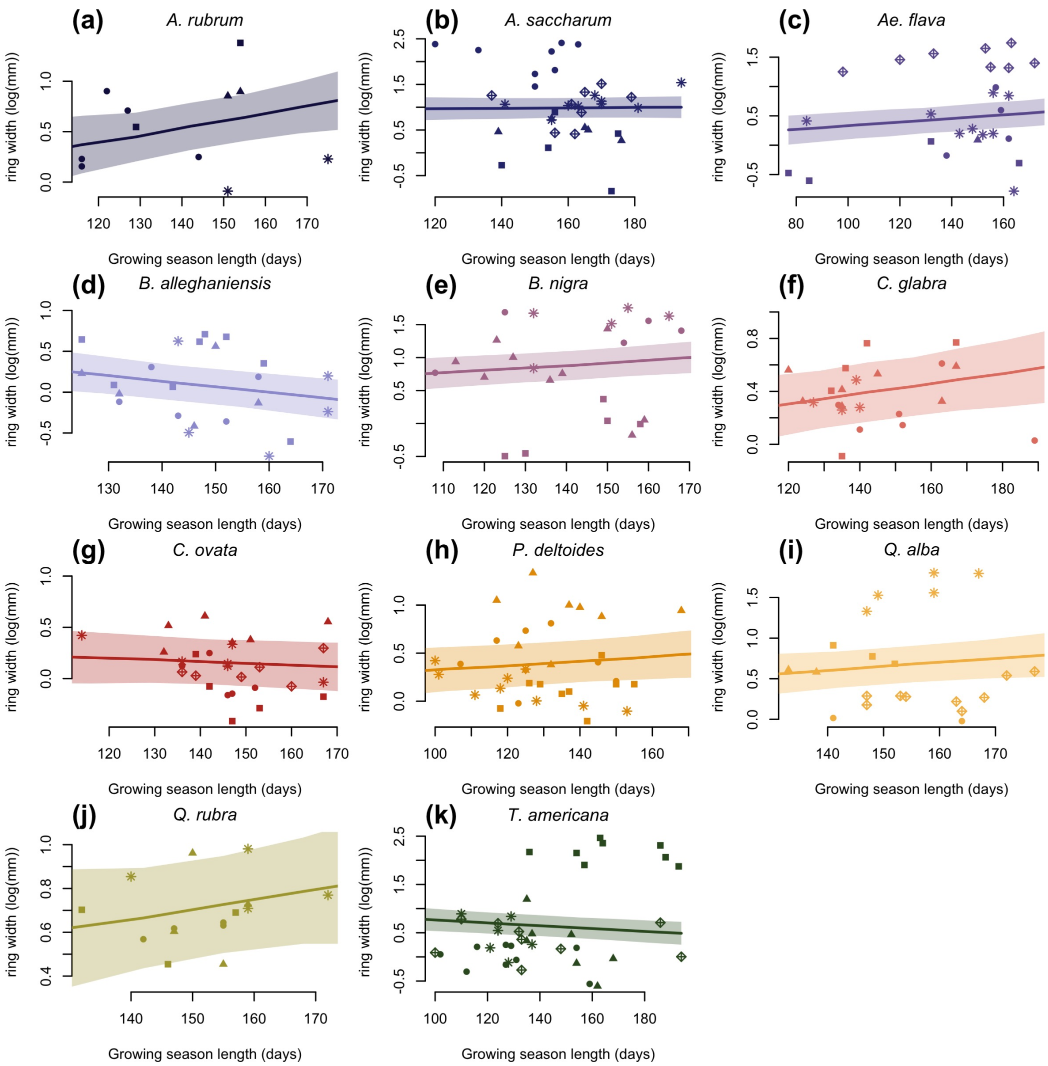

Figure S4: Ring width trends (on the log scale) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

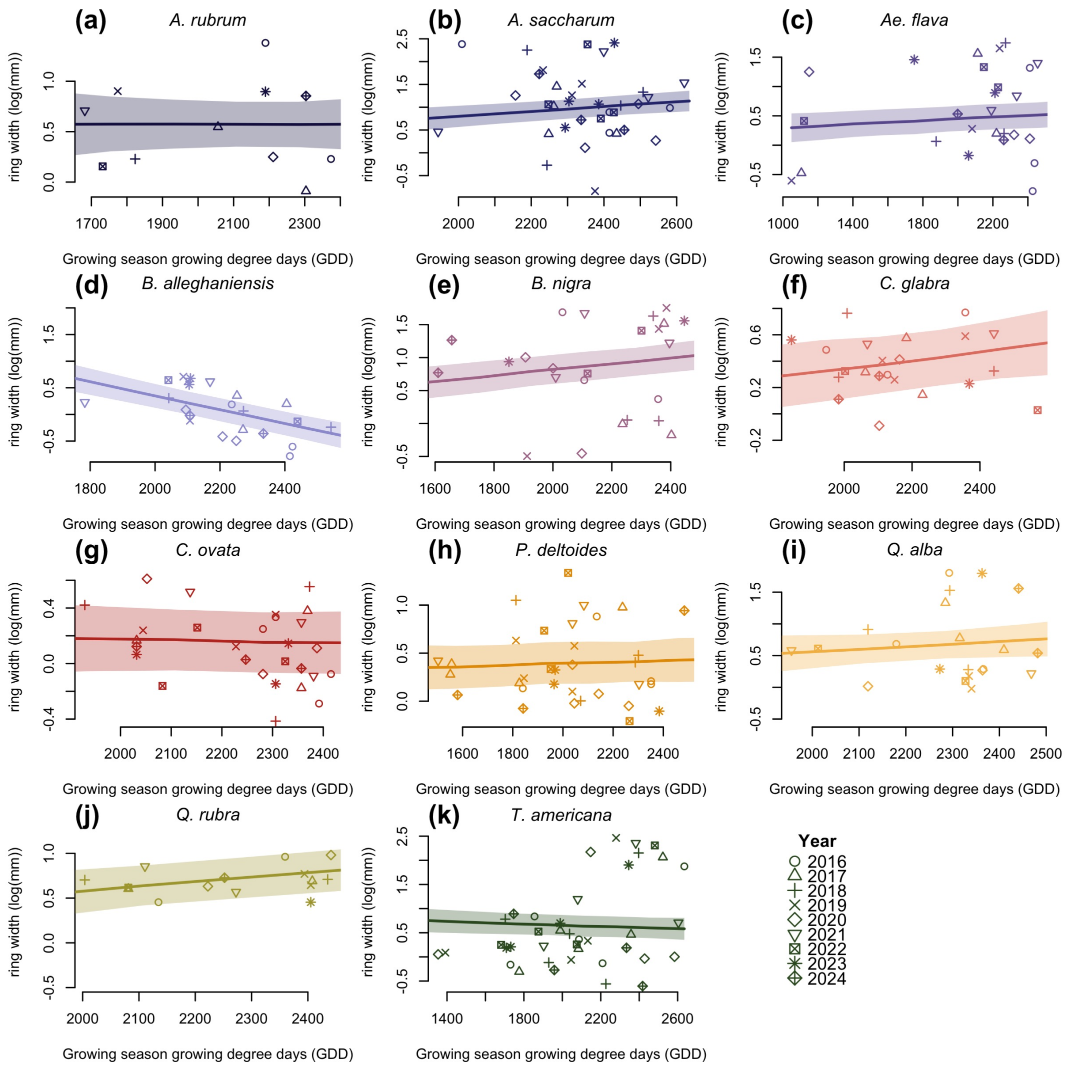

Figure S5: Ring width trends (on the log scale) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

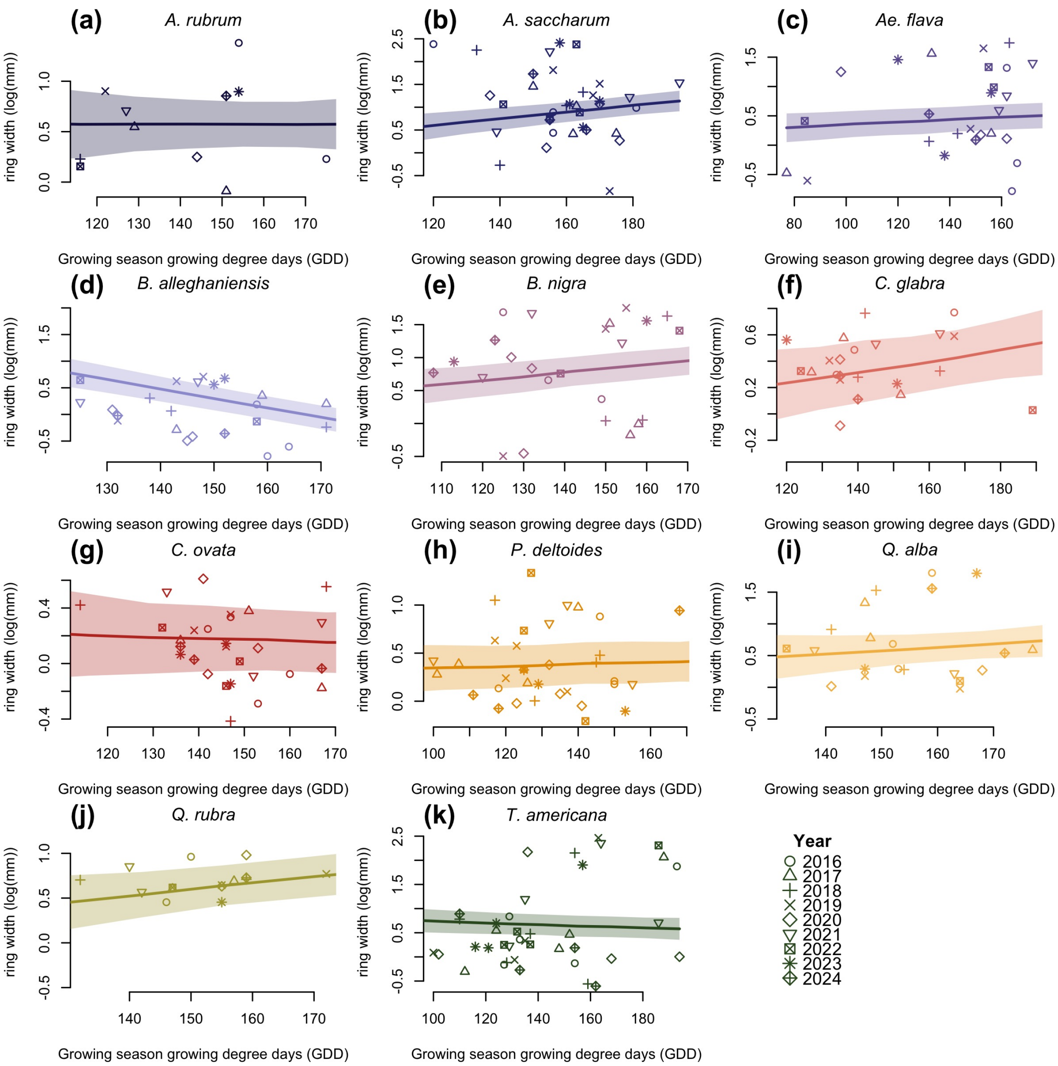

Figure S6: Ring width trends (on the log scale) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

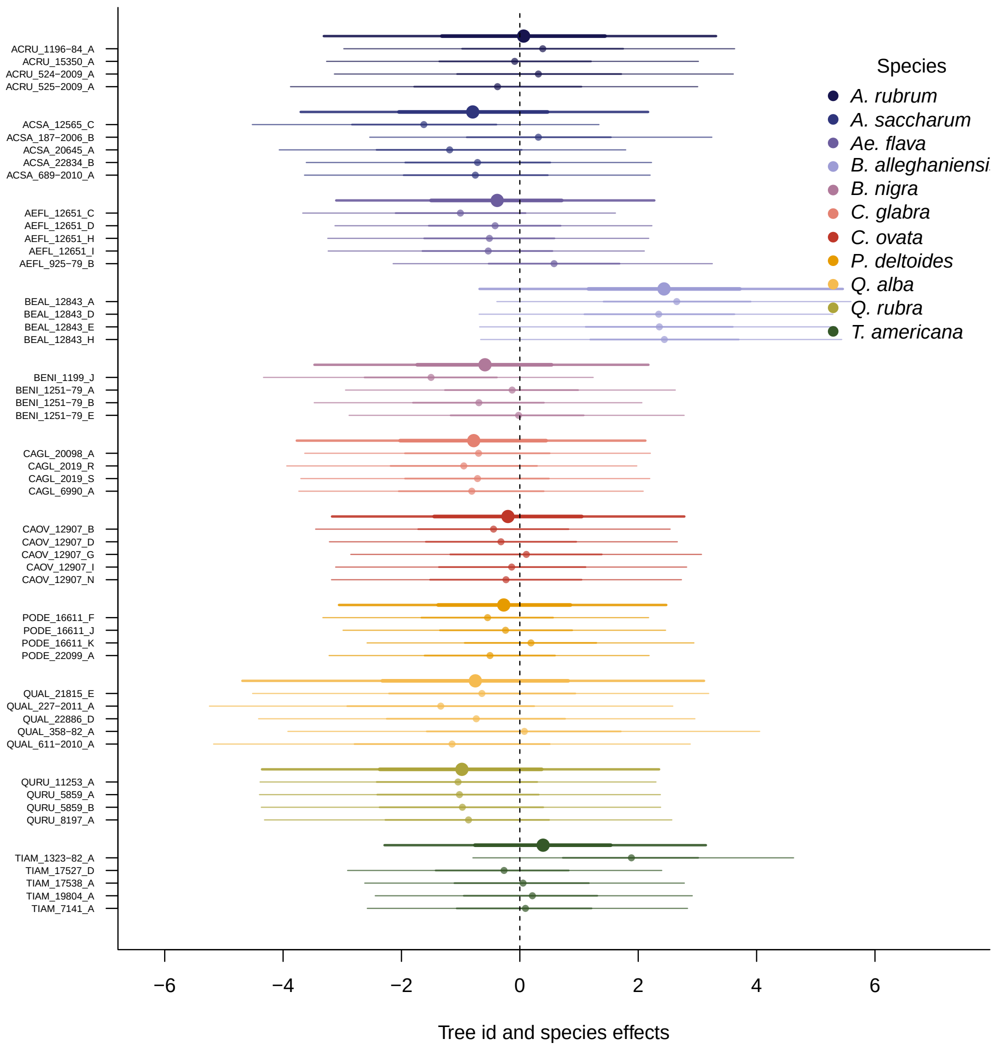

Figure S7: Ring width (on the log scale) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

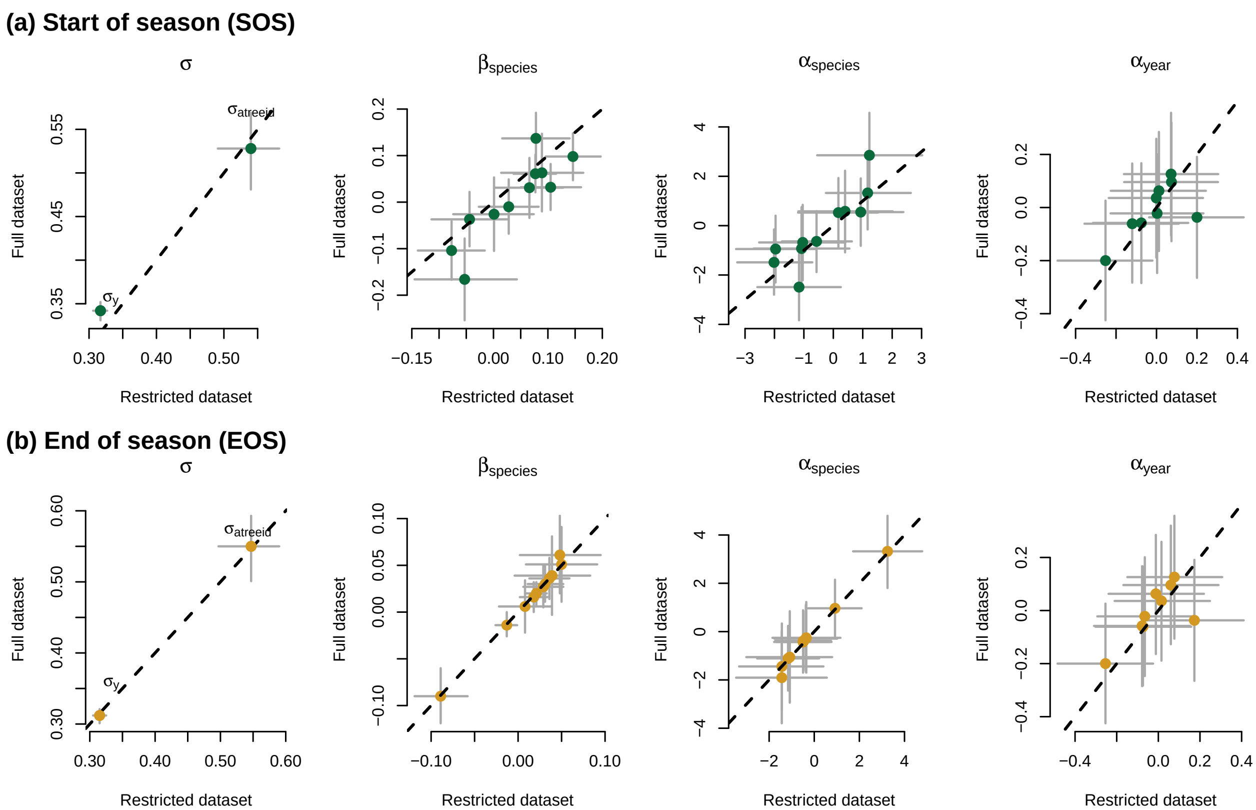

Figure S8: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor. Each pannel represent the different parameters used to fit the models (see data analyses for more details).

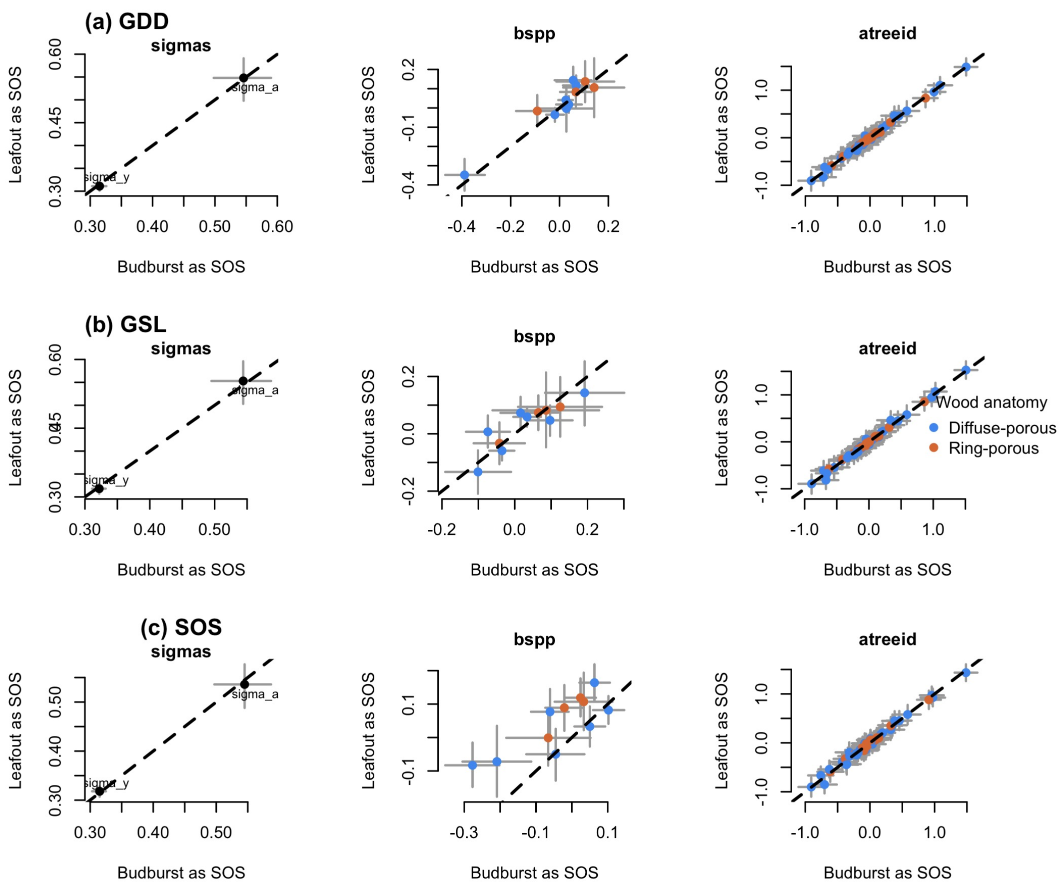

Figure S9: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see data analyses for more details).

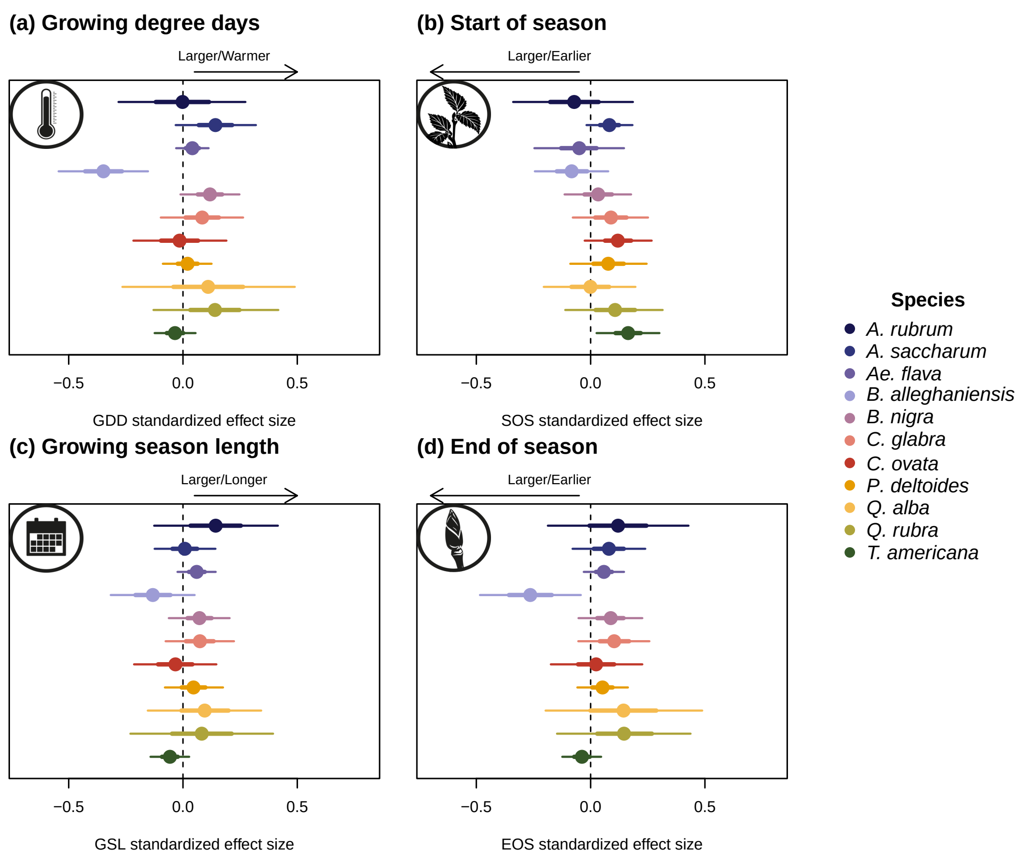

Figure S10: Effect size of each predictors across our eleven species. We depict ring width (on the log scale) change per scaled (z-scored; see data analyses section for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

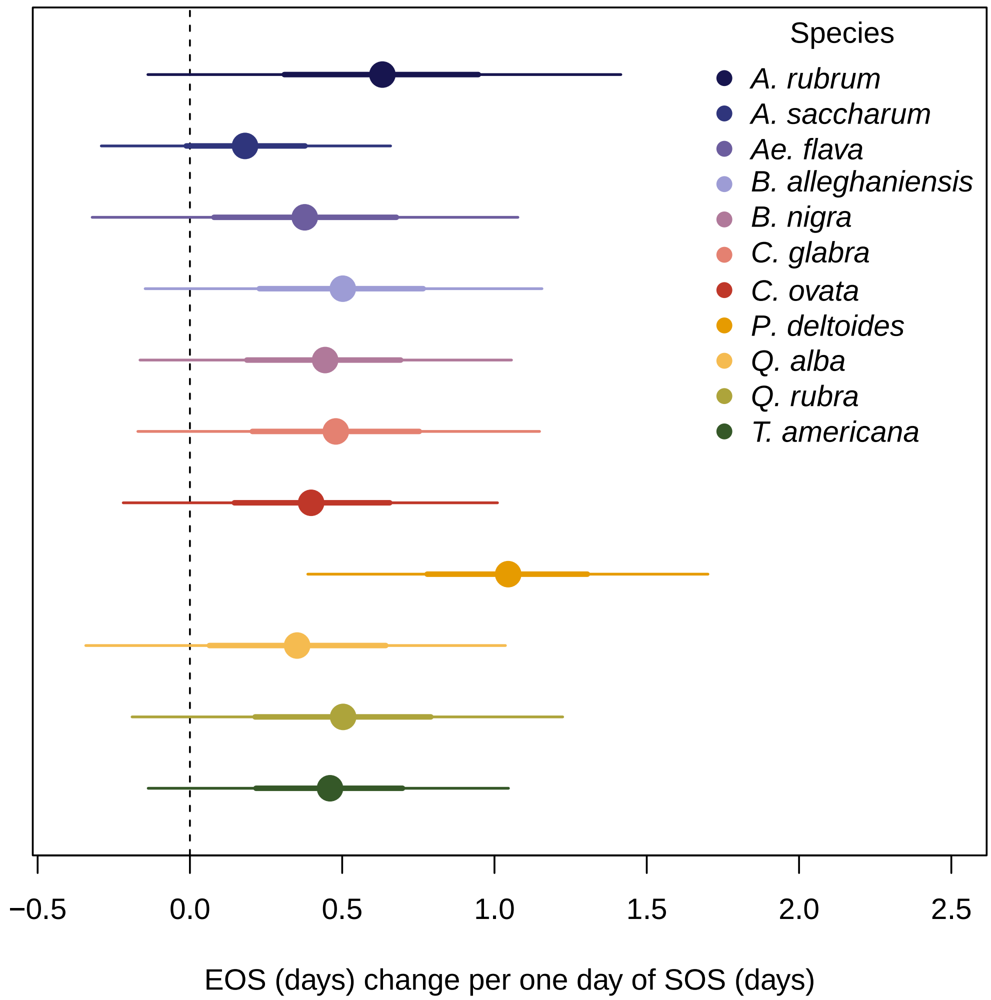

Figure S11: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.

References